week 8 09.03.2026

## Allocation of indivisible goods

① model:

- A set $G$ of $m$ indivisible goods
  - paintings, jewellery, house, car
- A set of players $N = \{1, \dots, n\}$
- valuations

$$v_i : 2^G \to \mathbb{R} \ (\mathbb{Z})$$

where $v_i(\phi) = 0$

- valuations are additive if for all $S \subseteq G$ and $i \in N$,

$$v_i(S) = \sum_{g \in S} v_i(g)$$

- An allocation is a partition of goods, $A = (A_1, \dots, A_n)$
  - $A_i \subseteq G$ is the bundle allocated to agent $i$
  - $A_i \cap A_j = \phi$ for $i \neq j$
  - allocation is complete if $\bigcup_{i \in N} A_i = G$ (and partial otherwise)

## ② Solution concepts:

An allocation $A = (A_1, \dots, A_n)$ is

- <u>proportional</u> if

$$v_i(A_i) \geq \frac{v_i(G)}{n} \quad \forall i \in N$$

- <u>envy-free</u> if

$$v_i(A_i) \geq v_i(A_j) \quad \forall i, j \in N$$

- <u>equitable</u> if

$$v_i(A_i) = v_j(A_j) \quad \forall i, j \in N$$

- <u>Pareto-optimal</u> if

there is no other allocation $B = (B_1, \dots, B_n)$ such that $v_i(B_i) \geq v_i(A_i)$ $\forall i \in N$ and $v_i(B_i) > v_i(A_i)$ for some $i \in N$

Unlike divisible settings like cake cutting, we cannot divide goods to obtain a fine-grained adjustment; indivisibility forces into a combinatorial, sometimes impossible trade-off between fairness and efficiency => envy-freeness and proportionality are generally infeasible.

<u>Example</u>: 2 agents, 1 item; both agents assign positive value to the item

## ③ Relaxations: PROP1 and EF1

An allocation $A = (A_1, \dots, A_n)$ is <u>proportional up to 1 good</u> (PROP1) if for each agent $i$ there is an item $g \in G$ s.t.

$$v_i(A_i \cup \{g\}) \geq \frac{v_i(G)}{n}$$

An allocation $A = (A_1, \dots, A_n)$ is <u>envy-free up to 1 good</u> (EF1) if for each pair of agents $i, j \in N$ there is an item $g \in A_j$ such that

$$v_i(A_i) \geq v_i(A_j \setminus \{g\})$$

<u>Example</u>: 2 agents, 1 item
- giving the item to one agent is PROP1 and EF1.

(4) <u>EF1 for additive valuations: Round-robin algorithm</u>

- agents take turns choosing their most preferred item among the remaining items
- in rounds $r = 1, 2, \dots$ player $(r \bmod n)$ picks their favourite remaining good

This round-robin (RR) allocation always ensures an EF1 allocation. The intuition is that player $i$ cannot envy player $j$ by more than one good, since if $j$ chose before $i$ in one round, $i$ will choose before $j$ in the next round. Within each phase, we will ensure EF1 allocation, and then since the valuations are additive, we can extend the claims across rounds (see proof in the video).

<u>Example</u>:

| | A | B | C | D |
|---|---|---|---|---|
| $v_1$ | 9 | 5 | 4 | 1 |
| $v_2$ | 6 | 7 | 5 | 2 |
| $v_3$ | 3 | 2 | 8 | 10 |

RR : $1 \to 2 \to 3 \to 1$

$A_1 = \{A, C\}$ $v_1 = 9 + 4 = 13$

$A_2 = \{B\}$ $v_2 = 7$

$A_3 = \{D\}$ $v_3 = 10$

However, RR allocation is not Pareto-efficient or strategyproof (see examples in the videos).

(5) EF1 for monotone valuations: envy-cycle elimination (ECE) algorithm

Valuations are monotone if for all $S \subseteq T \subseteq G$ and $i \in N$,

$$v_i(S) \leq v_i(T)$$

their [illegible]

## Envy graph

- vertices: agents
- there is an arc from $i$ to $j$ if $i$ envies $j$

## Reallocation along a cycle (RAC)

- suppose the envy graph contains a cycle
  $i \to j \to k \to \dots \to t \to i$
- reallocate the bundles along the cycle:
  $A_j \to i$
  $A_k \to j$
  ...
  $A_i \to t$
- we only permute the bundles, so the allocation remains EF1
- the cycle can be eliminated

# # of RAC steps

- we only reallocate the bundles but do not change them

- let all agents rank the $n$ bundles and score them, according to this ranking

- let the score of an allocation be the sum of the agents' scores for their bundles

- an RAC step
  - makes agents on the cycle happier
  - does not affect other agents
  
  $\Rightarrow$ increases the total score

- The total score ranges from 0 to $n(n-1)$ $\Rightarrow$ there are at most $n(n-1)$ steps

# ECE algorithm

- start with empty allocation (EF1).
- repeat until all items are allocated:
  - if there is an unenvied agent, allocate them an item
  - otherwise, there is an envy cycle
  - perform an RAC step (EF1)
- if no RAC step can be performed, there is no cycle, so there is an unenvied agent
- Polynomial runtime:
  - $m$ item allocation steps
  - $\leq n(n-1)$ RAC steps between any two allocation steps

$\boxed{\text{show ECE allocation on the same example}}$

As well as RR, ECE is also <u>not</u> Pareto-efficient (example in the videos).

<u>To summarise</u>: EF does not always exist; EF1 always exists for additive (RR) and monotone (ECE) valuations;

however, EF1 cannot be satisfied along with Pareto-optimality in the general case.

## (6) Concepts of welfare

- <u>utilitarian</u> welfare:

  sum of utilities

  $$UW(A) = \sum_{i \in N} v_i(A_i)$$

  always PO but may be highly unfair

- <u>egalitarian</u> welfare:

  minimal utility

  $$EW(A) = \min_{i \in N} v_i(A_i)$$

  neither PO nor EF1

- <u>Nash</u> welfare:

  product of utilities

  $$NW(A) = \prod_{i \in N} v_i(A_i)$$

  both PO and EF1

## ⑦ Envy-freeness up to any good (EFX)

An allocation $A = (A_1, \dots, A_n)$ is envy-free up to any good (EFX) if for each pair of agents $i, j \in N$ and for each item $g \in A_j$ we have

$$v_i(A_i) \geq v_i(A_j \setminus \{g\})$$

EFX is strictly stronger than EF1 and strictly weaker than perfect envy-freeness (EF).

- EFX exists for 2 agents with monotone valuations
- EFX exists for 3 agents with additive valuations
- For $n \geq 4$ agents — we don't know!

But it is guaranteed for identical additive valuations.

## (8) Maximin Share Guarantee

The maximin share (MMS) guarantee of player $i$ is given by:

$$\max_{X_1, \dots, X_n} \min_j v_i(X_j)$$

Intuitively, the MMS of player $i$ represents the maximum value they could guarantee themselves if they were to partition goods into $n$ bundles and then receive the worst bundle (i.e., "you cut, adversary chooses").

An MMS allocation ensures that $v_i(A_i) \geq$ MMS guarantee for all $i \in N$.

Intuition: think of MMS guarantee as answering the question: "what's the best I could do if I had to divide the items into n piles, knowing I'll get the least valuable pile?"

It captures a concept of fairness that's weaker than proportionality but can more often be achieved with indivisible goods.

Existence:

- For $n=2$, an MMS allocation always exists

- For $n \geq 3$, there exist additive valuation functions that do not admit an MMS allocation.

$\boxed{\text{The counterexample is non-trivial!}}$

The notion of MMS guarantees only came about in the 2010's and it took about 4 years to get to a counterexample for 3 agents. In practice, MMS guarantee is usually feasible, but it's not something that you can guarantee theoretically, as a hard fairness guarantee.

Later work has shown that we can always find allocations that provide each player at least $\frac{3}{4}$ of their MMS, and in many realistic settings, full MMS allocations do exist. This is a pattern that we often see: theoretical worst-case results might be negative, but practical instances tend to behave much better.

| | A | B | C | D |
|---|---|---|---|---|
| $v_1$ | 9 | 5 | 4 | 1 |
| $v_2$ | 6 | 7 | 5 | 2 |
| $v_3$ | 3 | 2 | 8 | 10 |

RR: $1 \to 2 \to 3$

$A_1 = \{A, C\}$ $\quad v_1 = 9 + 4 = 13$

$A_2 = \{B\}$ $\quad v_2 = 7$

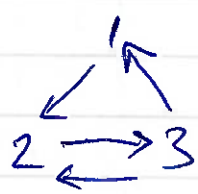
$A_3 = \{D\}$ $\quad v_3 = 10$

ECE: $D \to C \to B \to A$

$A_1 = \{D\}$

$A_2 = \{C\}$

$A_3 = \{B\}$




$A_1 = \{C\}$ $\quad A_1 = \{A, C\}$

$A_2 = \{B\}$ $\quad A_2 = \{B\}$

$A_3 = \{D\}$ $\quad A_3 = \{D\}$

max
Utilitarian welfare: allocate items to the agents that value them the most

$A_1 = \{A\}$ $v_1 = 9$
$A_2 = \{B\}$ $v_2 = 7$
$A_3 = \{C, D\}$ $v_3 = 18$

$\boxed{\Sigma = 34}$

max
Egalitarian welfare:

$A_1 = \{A\}$ $v_1 = 9$
$A_2 = \{B, C\}$ $v_2 = 7+5 = 12$
$A_3 = \{D\}$ $v_3 = 10$

$\boxed{min = 9}$

max
Nash welfare:

$A_1 = \{A\}$ $v_1 = 9$
$A_2 = \{B\}$ $v_2 = 7$
$A_3 = \{C, D\}$ $v_3 = 18$

$\boxed{\Pi = 1134}$

MMS: $M_1 = \min\{9, 5, 4+1\} = 5$
$M_2 = \min\{6, 7, 5+2\} = 6$
$M_3 = \min\{3+2, 8, 10\} = 5$

MMS allocation: any of the above